Architecture Justification

Smart Lab Equipment & Slot Reservation Portal — Epic 1: Reserve Equipment Slot (SEL2-1)

Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal. The system is organised into a Presentation Layer (Reservation UI), a Business Layer (Reservation Manager, Availability Checker, Calibration Verifier, Slot Lock Manager, Token Generator) and a Data Layer (Database Component), communicating through defined provided/required interfaces.

Architectural Style Analysis

Style	Pros	Cons	Fit for this portal
Layered	Clear separation of concerns; simple to build and test	Some layer overhead; harder to scale single parts	**Best fit** – matches the sequential reservation workflow
Microservices	Independent scaling and deployment; fault isolation	Operational complexity; network latency; data consistency issues	Over-engineered for one lab and a small student team
Client-Server	Centralised control; easy data consistency	Single point of failure; scalability bottleneck; no internal structure	Too coarse; does not separate validation, locking and token logic

Two Reasons for the Choice

1. The reservation workflow is a sequential pipeline that maps naturally onto layers. In Lab 2, the six user stories follow a fixed order: view availability (SEL2-2), check slot availability (SEL2-3), verify calibration (SEL2-4), enforce reservation limits (SEL2-5), lock the slot (SEL2-6) and generate the token (SEL2-7). The Reservation UI only collects input and shows results, the Business Layer applies these rules in order, and the Data Layer stores equipment, slots and reservations. This also mirrors our sprint plan: Sprint 1 (14 pts) built the validation checks and Sprint 2 (8 pts) the lock and token steps, so each layer could be built and tested independently within one-week sprints at our 14–16 point capacity.

2. Concurrent bookings need one consistent source of truth. Epic 1 (FR-001 to FR-003, NFR-001, NFR-002) requires preventing overlapping reservations and processing simultaneous requests safely, so two students can never hold the same slot. A single shared Data Layer lets the Slot Lock Manager use one database transaction to lock a slot, avoiding the distributed data-consistency problems of microservices. The portal serves one lab's students, so independent scaling of services is not needed.

Security Advantage

Students never reach the data directly. Every request passes from the Presentation Layer through the Business Layer, where authentication, calibration status, duration and advance-booking limits are validated on the server before any reservation is written. The Database Component is exposed only through the Business Layer's interface, which reduces the attack surface and keeps confirmed records and unique reservation tokens protected from tampering or bypassed rules.

Performance Benefit

All layers run in the same application, so calls between the Reservation Manager, Slot Lock Manager and Database Component are in-process with no network hops. A slot check and lock completes in a single short transaction on indexed equipment/slot tables, giving fast availability checks and reservation confirmation even when many students book at once, unlike microservices where each step would add network latency.

Identified Components and Key Interfaces

Component	Layer	Interface (provided / required)
Reservation UI	Presentation	Requires IReservation (API calls)
Reservation Manager	Business	Provides IReservation; requires IAvailability, ICalibration, ISlotLock, IToken
Availability & Limits Checker	Business	Provides IAvailability; requires IData (database queries)
Calibration Verifier	Business	Provides ICalibration; requires IData
Slot Lock Manager	Business	Provides ISlotLock; requires IData (transactions)
Token Generator	Business	Provides IToken; requires IData
Database Component	Data	Provides IData (SQL queries)